Jiacheng Wang

+33 7 64 43 53 85 | jiacheng.wang@eurecom.fr | Paris, France
github.com/jiacwng | linkedin.com/in/jiacheng-wang

Summary

Final-year Digital Security engineering student at EURECOM (Mines-Télécom), seeking a 6-month international internship, based outside France, from September 2026. Focused on SOC alert triage, static and dynamic malware analysis, reverse engineering, and machine learning for security, and eager to put it into practice.

Projects

Meerkat | ML-Assisted SOC Alert Triage | Jul 2026

github.com/jiacwng/meerkat

- Built a Python CLI for Tier 1 SOC alert triage, normalizing Wazuh, Suricata and AMiner alerts into one MITRE ATT&CK-enriched schema.
- Grouped repeated alerts per host and rule into a budget-limited review queue, maximising distinct attack activity covered per analyst decision.
- Scored candidates with a Random Forest and logistic re-ranker, surfacing 58 of 60 attack steps in a ten-case daily budget against 33 for native detector severity, on the AIT simulated-company dataset.
- Shipped ATT&CK Navigator export, analyst decision capture, and model-drift detection via Population Stability Index.

eous | Fuzzy Hashing for PE and ELF Binaries | May 2026 - Present

github.com/jiacwng/eous

- Built eous, a fuzzy-hashing tool for PE and ELF binaries that matches the same program across compilers with 97% top-1 accuracy, against 69% for the industry-standard tlsh.
- Designed a 512-bit MinHash over disassembled instruction 6-grams (LIEF, iced-x86) that survives the byte and address rewrites of recompilation, benchmarked on 599 binaries across 26 programs.
- Added six quality gates that decline unreadable input (.NET, packed, no code, unsupported) with a named reason instead of a false match.

SmudgeC | Native C Source-to-Source Obfuscator | Jun 2026 - Present

github.com/jiacwng/smudgeC

- Built a dependency-free C source-to-source obfuscator in pure C, with a hand-built lexical scanner and symbol table.
- Implemented configurable passes (identifier renaming, comment stripping, literal encoding, minification, runtime string hiding that removes plaintext from the binary), preserving C keywords, standard-library, macro and local-header names via protected-name lists.

CTF Writeups | Binary Exploitation, Web, Forensics | 2026 - Present

github.com/jiacwng/ctf-writeups

- Authored 25+ walkthroughs and exploit chains for picoCTF challenges, demonstrating practical vulnerability exploitation and penetration-testing techniques.

Academic Projects

Mnemocrypt | Cryptographic Function Detection in x86 Binaries | Nov 2025 - Feb 2026

EURECOM semester project, extending an IDA Pro plugin from published research

- Broadened the training data beyond cryptographic libraries with a curated 13-library goodware corpus, embedded cryptographic routines stripped from the negative class.
- Cut false positives on 1,481 held-out functions by 81% (48 to 9) and raised precision from 0.60 to 0.86 with lane-aware SIMD instruction counting.
- Prototyped operand-level features (AES S-box, SHA-256 constants); excluded them after weak transfer to real malware.

Experience

Software Engineering Intern | Reva 2, Nice, France | Jul 2025 - Sep 2025

- Contributed to a secure full-stack web application in React and Express.js.
- Implemented defence-in-depth controls: Helmet security headers, strict CORS policies, API rate limiting and environment-based secrets management.

Education

EURECOM, Mines-Télécom | Sophia Antipolis, France | Expected Mar 2027

French Engineering degree (Diplôme d'ingénieur) specialized in Digital Security

- *Core modules:* System and Network Security, Hardware Security, Digital Forensics, Secure Communications, Machine Learning and Intelligent Systems, Computer Architecture, Distributed Systems and Cloud Computing.

Classe Préparatoire (CPGE), Lycée Descartes | Tours, France | Sep 2021 - Jul 2023

MPI track (mathematics, physics and computer science), the selective national route into French engineering schools, concluding with competitive entrance examinations.

Skills

Programming: C (C99), Python, OCaml, Java, JavaScript, SQL, MATLAB; assembly (x86-64, RISC-V).
Security & Reverse Engineering: IDA Pro, Ghidra, gdb, Capstone, LIEF, Wazuh (SIEM), Suricata, Wireshark, nmap, MITRE ATT&CK.
Machine Learning: scikit-learn, pandas, NumPy.
Systems & Tools: Linux, Git, GitHub Actions, pytest, VirtualBox, Open5GS (5G core lab), Microsoft Azure.
Spoken Languages: Mandarin (Native), French (Native), English (TOEIC 930).

Certifications

Handimanage Label | Companieros | Oct 2025 - Jan 2026

Certified in managing and promoting disability inclusion in the workplace.

Volunteering

Volunteer French Teacher | Amours et Coeurs Unis en France | Mar 2026 - Present

Over 100 hours teaching French to non-native speakers and assisting them with administrative procedures.

Treasurer | School Student Association | 2023 - 2024

Managed the annual budget for one year, funding student events and reporting on spending.